Chapter 6
Interest Rate Futures

Interest rate futures are the most heavily traded derivatives in existence, and they are where most of the hedging that the rest of this book describes actually gets done. A dealer who has warehoused the risk of a swap, a structured note, or a mortgage portfolio does not usually offload it into another bespoke trade; it hedges in the futures market, where the liquidity is.

Two families dominate. **Short-term interest rate futures** settle against an overnight rate averaged over a reference period, and are used to trade and hedge the front of the curve. **Treasury futures** require physical delivery of a government bond, and are used further out. The two are constructed on completely different principles, and each has a wrinkle that repays attention: the short-term contracts carry the convexity adjustment of Chapter 30, and the Treasury contracts embed a delivery option that decides which bond the contract tracks.

The chapter closes with a trade that arises directly from that second wrinkle: the cash–futures basis trade, which hedge funds have built to the high hundreds of billions of dollars, and which in March 2020 took roughly $1 trillion of Federal Reserve purchases to arrest. It is a recurring subject of financial stability reports for that reason.

Section 6.1 sets out the quotation conventions without which none of the prices can be read. Section 6.2 covers the short-term contract, and Sections 6.3 and 6.4 the Treasury contract and the delivery option that decides which bond it tracks. Section 6.5 turns to the basis trade built on that option, and Sections 6.6 to 6.8 to the uses: hedging a swap book with the strip, duration-based hedging, and building the front of the curve from futures prices.

6.1 DAY COUNTS AND QUOTATION CONVENTIONS

Chapter 4 noted that a rate is meaningless without its day count. Futures markets add quotation conventions on top, and the combination is the subject's main source of confusion.

Day counts. US Treasury bonds accrue on actual/actual; corporate and municipal bonds on 30/360; money market instruments including SOFR on actual/360. The 30/360 convention treats every month as thirty days and gives no month a 31st, which produces the small anomaly that a bond accrues interest on February 29 and 30 and none on the 31st of a long month.

Price quotation for bonds. US Treasury bonds and notes are quoted in points and thirty-seconds of a point, per $100 of face value. A quote of `110-16` means $110 + 16/32 = 110.50$. Finer increments appear as a third digit: `110-165` means $110 + 16.5/32 = 110.5156$, and the shorter note futures are quoted in quarters of a thirty-second.

Clean and dirty price. A coupon is paid in full to whoever holds the bond on the payment date, although a seller may have held it for most of the period. The seller is compensated through **accrued interest**, the part of the next coupon earned since the last coupon date on the

bond's day count:

$$\text{accrued interest} = \text{coupon} \times \frac{\text{days since the last coupon}}{\text{days in the coupon period}}$$

The buyer pays it to the seller on top of the price and receives the whole coupon back on the payment date. The quoted price is the **clean price**, which excludes accrued interest. The amount actually paid is the **dirty price**, or cash price:

$$\text{cash price} = \text{quoted price} + \text{accrued interest since the last coupon} \tag{6.1}$$

A 4% Treasury paying coupons on March 1 and September 1, bought at `101-08` for June 15 settlement, has accrued 106 of its 184-day coupon period on actual/actual, so per $100 of face the buyer pays

$$101.25 + 2.00 \times \frac{106}{184} = 101.25 + 1.1522 = 102.4022$$

The convention exists so that a bond's quoted price does not sawtooth upward between coupons and drop on each payment date, which would obscure genuine price movement.

Price quotation for short-term rate futures. Short-term interest rate futures are quoted as

$$\text{quoted price} = 100 - R \tag{6.2}$$

where R is the rate in percent. A quote of 95.72 corresponds to a rate of 4.28%. The convention inverts the usual direction: a trader who expects rates to *rise sells* the futures contract. It makes the contract behave like a price, rising when the holder profits, as price-based margining systems require.

6.2 SOFR FUTURES

The Eurodollar futures contract, which settled against three-month LIBOR, was for decades the most actively traded futures contract in the world. Its replacement settles against SOFR, and the difference in construction has consequences.

Case Study 6.1 Retiring the largest futures contract ever traded

The Eurodollar contract launched in 1981 and became the benchmark for the front end of the dollar curve, with open interest at times above ten million contracts, over $10 trillion of notional.

It could not simply be delisted: millions of open positions, some running for years, referenced a rate that would cease to exist. CME's solution, executed in April 2023, was to convert open Eurodollar positions into the corresponding SOFR futures, applying the same ISDA fallback spread adjustment that Chapter 4 described for swaps (26.161 basis points for the three-month tenor), so that the economics of each position carried over.

The conversion was, by the standards of such things, uneventful. That is the point worth noting: the transition succeeded because the compensation mechanism was fixed in advance and applied uniformly, so that no participant had to negotiate. Where fallback language was ambiguous, disputes followed instead.

Contract construction

Two contracts cover the front end.

Three-month SOFR futures have a notional of $1,000,000 and settle against the *compounded* daily SOFR over a three-month reference period running between consecutive

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International Money Market (IMM) dates (the third Wednesdays of March, June, September, and December). Since a basis point on \$1,000,000 for three months is

$$\$1{,}000{,}000 \times 0.0001 \times 0.25 = \$25$$

each basis point is worth \$25 per contract; Example 6.1 follows a position to its profit.

One-month SOFR futures have a notional of \$5,000,000 and settle against the *arithmetic* average of daily SOFR over a calendar month. A basis point is worth

$$\$5{,}000{,}000 \times 0.0001 \times \frac{1}{12} = \$41.67$$

Example 6.1 Profit on a SOFR futures position

A trader expects the central bank to tighten more than the market has priced, and sells 200 three-month SOFR futures at 95.72, an implied rate of 4.28%.

Rates subsequently rise 12 basis points and the contract settles at 95.60, an implied rate of 4.40%. Because the trader is short the price, the gain is

$$200 \text{ contracts} \times 12 \text{ bp} \times \$25 = \$60{,}000$$

The trader profited from rates rising while being *short* the contract. The initial margin required would have been a small fraction of the \$200 million notional involved, which is what makes futures the instrument of choice for expressing rate views.

The averaging feature and delta decay

These contracts have a property with no counterpart in the Eurodollar world, one that Sections 29.4 and 30.3 return to, and it is worth seeing directly.

A Eurodollar contract settled against a rate fixed on a single date. Until that date its sensitivity to rates was full; after it, the contract was over. A SOFR contract settles against an average accumulating across its reference period. Each day that passes locks in one more daily fixing, so the contract's remaining sensitivity falls, approximately linearly, from full at the start of the reference period to zero at its end. Rates desks call a first-order sensitivity of this kind a **delta**, which is why the effect is known as *delta decay*; the same word names an option's sensitivity to its underlying in Chapter 18, and the two are the same idea applied to different variables. Figure 6.1 contrasts the two contracts.

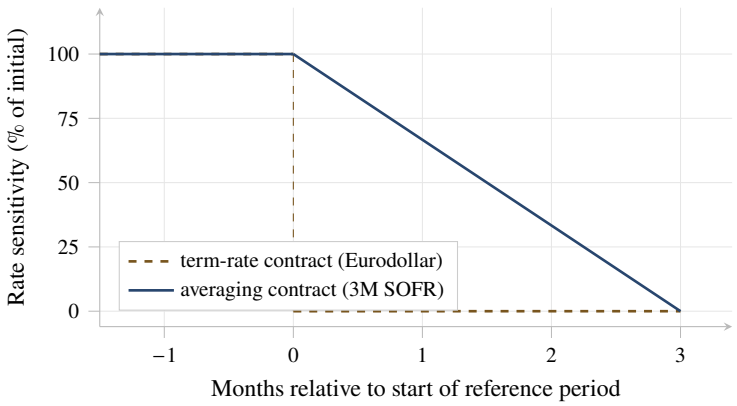


Figure 6.1: Rate sensitivity through the reference period.

The practical consequence is that a hedge built from SOFR futures must be rebalanced *through* the reference period rather than rolled at a single date. A position that is correctly

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hedged on the first day of the quarter is under-hedged by the last, because the futures leg has lost sensitivity that the hedged item has not.

Who uses them, and for what

The contracts are rarely traded one at a time. A **strip** is a set of contracts of the same type spanning a run of consecutive dates (or, in the option markets of Chapter 22, a run of strikes) bought or sold together and priced as a single position. The SOFR strip is the sequence of quarterly contracts stretching out along the curve, and most of what follows is about the strip rather than about any one contract in it.

Four groups trade the SOFR strip, and their separate motives explain the contract's liquidity.

Borrowers and lenders hedging near-term funding. A corporate treasurer expecting to roll commercial paper (short-term unsecured company borrowing) in six months, or a bank funding a loan book at short rates, locks in the cost by selling futures for the relevant quarter. This is the textbook hedge and the contract's original purpose.

Dealers hedging swap books. This group is by far the largest source of volume. A dealer that has received fixed on a five-year swap holds front-end delta it does not want, and offloads it by selling a strip of SOFR futures covering the swap's early accrual periods. Chapter 7 describes the residual risks this leaves; the futures market is where the first-order risk goes.

Traders positioning on monetary policy. The strip of consecutive contracts is a direct reading of the market's expected path for the policy rate, so a view that the central bank will move faster or slower than priced is expressed by buying one contract and selling another. Trades of this kind (the **steepener**, which profits if later rates rise relative to earlier ones, and the **flattener**, which profits if they fall) are among the most common macro trades.

Curve builders. The futures strip supplies the market's most liquid information about the forward rates of Section 4.9 between roughly three months and five years, and is a primary input to the curve construction of Chapter 7, after the convexity correction described next.

Futures rates are not forward rates

The final property, developed at length in Section 30.3, must be flagged here because it governs how these contracts are used to build a curve. Daily settlement favors the short side of the contract, which gains when rates rise: its gains arrive when rates are high and can be reinvested at those rates, and its losses arrive when rates are low and are cheap to finance. A forward rate agreement, settled once, offers neither side that advantage, so the futures rate must sit *above* the forward rate to compensate the long. The gap is the **convexity adjustment**. How large it is depends on the model used for the short rate; the simplest form, and the one this chapter quotes, comes from the Ho–Lee model of Chapter 31, in which the short rate moves by normally distributed amounts with constant volatility:

$$\text{Forward rate} = \text{Futures rate} - \frac{1}{2}\sigma^2 T_1 T_2 \tag{6.3}$$

where T_1 is the time until the contract's rate is determined, T_2 the time until the end of the period that rate covers, and σ the standard deviation of the absolute change in the short rate over a year (a normal volatility, which is the market convention for this parameter). Strictly, (6.3) adjusts the continuously compounded rate; Section 30.3 adds the small term for the simply compounded rate the contract settles on. At $\sigma = 1.00\%$ the adjustment is under a basis point on the one-year contract and about 51 basis points on the ten-year, growing with the *square* of maturity—though Ho–Lee has no mean reversion, no pull of the rate back toward a long-run level, and Section 30.3 shows that this overstates the long end once mean reversion is allowed for. Curve construction that treats quoted futures rates as forward rates will be visibly wrong beyond the first two years or so.

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6.3 TREASURY FUTURES

Treasury futures rest on a different principle: physical delivery of a government bond.

Two features of that delivery shape what follows, and both are the short's choice. *Delivery is not a date but a month.* Any business day of the contract's delivery month will serve, and it is the **short** who declares which one, by giving notice; the long takes delivery when told. *And the short chooses the bond as well*, from a basket set by the exchange. Neither choice is a technicality; together they are worth a material amount, priced in Section 6.4.

Physical delivery raises a further difficulty: no single bond stays suitable. A contract referencing one specific ten-year note would become a contract on a nine-year note within a year, and would be vulnerable to a squeeze if someone accumulated the issue. The exchanges' solution is to define a **deliverable basket**: the short may deliver any bond satisfying a maturity range on the delivery date. Table 6.1 lists the principal US contracts.

Table 6.1: Principal US Treasury futures contracts.

Contract	Approximate deliverable maturity	Face value
2-year note	1¾–2 years	\$200,000
5-year note	4¼–5¼ years	\$100,000
10-year note	6½–10 years	\$100,000
Treasury bond	15–25 years	\$100,000
Ultra bond	25 years and over	\$100,000

Conversion factors

Allowing any bond in the basket to be delivered creates an obvious problem: a high-coupon bond is worth more than a low-coupon one, so the short would always deliver the lowest-coupon bond, and the contract would track that one bond alone.

The remedy is the **conversion factor**. Each deliverable bond is assigned a factor equal to its price per \$1 of face value if it were priced to yield a notional standard rate (6% in the US contracts), so that a bond paying more than 6% is scaled up and one paying less is scaled down. The amount the long pays on delivery, writing CF for the conversion factor, is

$$\text{invoice} = (\text{settlement price} \times \text{CF} + \text{accrued interest}) \times \frac{\text{face}}{100} \tag{6.4}$$

Example 6.2 computes it for one delivery.

Example 6.2 The invoice amount on delivery

A ten-year note futures contract settles at `110-16`, that is 110.50. The short delivers a bond with conversion factor 0.8456 on which accrued interest is \$1.20 per \$100 of face; the contract covers \$100,000.

From (6.4),

$$(110.50 \times 0.8456 + 1.20) \times 1{,}000 = (93.4388 + 1.20) \times 1{,}000 = \$94{,}638.80$$

The factor scales the payment down because the delivered bond's coupon is below the 6% standard.

Who uses them, and for what

Treasury futures are the principal tool for moving duration around, and the identity of the users matters because it explains a persistent feature of the price.

Asset managers adjusting duration. A bond fund that wants to shorten duration ahead of an expected sell-off can sell futures rather than sell bonds: faster, cheaper in transaction costs,

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and without disturbing a carefully constructed credit portfolio. A fund that wants *more* duration than its cash allows buys futures, obtaining the exposure for margin rather than the full purchase price. This second use is important: it means a large and fairly price-insensitive stream of buying interest sits in the futures market, which tends to leave futures *rich* (expensive) relative to the cash bond. That richness is precisely what the basis trade of Section 6.5 harvests.

Dealers hedging inventory. A dealer holding bonds bought from customers hedges the price risk with futures until the position is distributed. Because the hedge is against a basket rather than the specific bond, what remains is basis risk, the subject of Section 6.4.

Mortgage portfolios. Holders of mortgage-backed securities face duration that *shortens* when rates fall, as borrowers prepay, and lengthens when rates rise, the property called **negative convexity**, since the price gains less in a rally than it loses in a sell-off. Keeping such a portfolio hedged requires constant rebalancing in the same direction as the market is already moving, and Treasury futures are the usual instrument. This is a well-known amplifier of rate moves.

Curve and spread traders. Buying one contract and selling another expresses a view on the shape of the curve (long two-year against short ten-year futures is a steepener) without an outright duration position. Trading futures against the bonds expresses a view on the basis itself.

Liability-driven investors. Pension funds and insurers with long-dated liabilities use long-maturity contracts to lengthen asset duration toward liability duration. Section 8.9 works through what happened in 2022 when this exposure was held with leverage.

How conversion factors are calculated

The definition is operational, and worth carrying out, because the two conventions buried in it are what make conversion factors imperfect.

Step 1: round the maturity. Measure the bond's remaining life from the first day of the delivery month and round it *down* to the nearest three months. A bond with 8 years and 5 months remaining is treated as having 8 years and 3 months; one with 8 years and 2 months is treated as having exactly 8 years. This rounding is what allows a single factor to be published for the whole delivery month rather than recomputed daily.

Step 2: price the bond at 6%. Discount the bond's coupons and principal at exactly 6% per annum, semiannually compounded, using the rounded maturity and \$1 of face; the result is the conversion factor.

For a bond whose rounded maturity is a whole number of half-years, this is an ordinary bond price calculation.

Example 6.3 Conversion factor for a bond with 8 years remaining

A bond pays a 5% annual coupon semiannually and has exactly 8 years remaining after rounding: $n = 16$ half-years, a coupon of \$2.50 per \$100 of face, and a discount rate of 3% per half-year.

The annuity factor is

$$a = \frac{1 - 1.03^{-16}}{0.03} = \frac{1 - 0.623167}{0.03} = 12.5611$$

so the price per \$100 of face is

$$2.50 \times 12.5611 + \frac{100}{1.03^{16}} = 31.4027 + 62.3167 = 93.7194$$

The conversion factor is this divided by 100:

$$\text{CF} = 0.9372$$

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It is below 1 because the bond's 5% coupon is below the 6% standard.

Now take an odd quarter. Suppose the same bond has 8 years and 3 months remaining after rounding. Three months from now it will have exactly 8 years to run and will just have paid a coupon, so at that date it is worth the \$93.7194 above plus the \$2.50 coupon:

$$93.7194 + 2.50 = \$96.2194$$

Discount that back one quarter at 3% per half-year. Today the bond is already three months into a six-month coupon period, and the interest accrued since the last (notional) coupon date is paid separately by the buyer, as accrued interest, so it must not be charged for twice; deduct it:

$$\frac{96.2194}{1.03^{1/2}} - 2.50 \times \frac{3}{6} = 94.8078 - 1.25 = 93.5578 \qquad \Longrightarrow \qquad \text{CF} = 0.9356$$

The factor is slightly below the 0.9372 of the whole-half-year case, because the extra quarter carries a below-coupon bond a little further from par. The general formula below is built for this case.

The general formula. When the rounded maturity ends on an odd quarter, the bond is valued at the next coupon date, three months away, discounted back, and reduced by the interest accrued since. The exchange publishes a single formula covering both cases. Writing c for the annual coupon rate, n for the number of whole years, and z for the remaining months ($z \in \{0, 3, 6, 9\}$), set $v = z$ if $z < 7$ and $v = z - 6$ otherwise. Then

$$\text{CF} = \frac{1}{1.03^{\,v/6}} \left[\frac{c}{2} + \kappa + \frac{c}{0.06}\,(1 - \kappa) \right] - \frac{c}{2} \cdot \frac{6 - v}{6} \tag{6.5}$$

where $\kappa = 1.03^{-2n}$ if $z < 7$ and $\kappa = 1.03^{-(2n+1)}$ otherwise. The final term deducts the interest accrued over the $6 - v$ months since the last coupon; the leading factor discounts back over the odd quarter. Example 6.4 checks the formula against Example 6.3.

Example 6.4 Applying the formula

Check against Example 6.3. Here $c = 0.05$, $n = 8$, $z = 0$, so $v = 0$ and $\kappa = 1.03^{-16} = 0.623167$:

$$\text{CF} = 1 \times \left[0.025 + 0.623167 + \frac{0.05}{0.06} \times 0.376833 \right] - 0.025 = 0.962194 - 0.025 = 0.9372$$

which reproduces the direct calculation exactly.

Now raise the coupon. With $c = 0.07$ and the same maturity,

$$\text{CF} = \left[0.035 + 0.623167 + \frac{0.07}{0.06} \times 0.376833 \right] - 0.035 = 1.0628$$

Above 1, as it must be for a coupon above 6%.

Finally, take an odd quarter. With $c = 0.05$, $n = 8$, $z = 3$, so $v = 3$ and κ is unchanged at 0.623167:

$$\text{CF} = \frac{1}{1.03^{0.5}}\,[0.962194] - 0.025 \times \frac{3}{6} = 0.985329 \times 0.962194 - 0.0125 = 0.9356$$

reproducing the direct construction at the end of Example 6.3. The bracket is that example's \$96.2194 per \$100, the leading factor is its discount back over the odd quarter, and the final term is its accrued deduction. The formula is that construction written out once.

Coupon	Rounded maturity	Conversion factor
5%	8 years	0.9372
7%	8 years	1.0628
5%	8 years 3 months	0.9356

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Why the factors do not equalize perfectly. Two approximations are now visible. The factors are computed at a *fixed* 6% yield, so they correctly equalize the basket only when the curve is flat at 6%; at any other level the scaling is systematically off, and increasingly so the further the curve sits from 6%. And the maturity rounding discards up to three months of a bond's life, which distorts factors slightly and in a direction that varies across the basket. Both errors are small but not zero, and the residue decides which bond is cheapest to deliver.

6.4 THE CHEAPEST-TO-DELIVER BOND

Because conversion factors equalize the basket only approximately, some bonds end up cheaper to deliver than others. The short, who chooses, will deliver whichever is cheapest—and since the contract is then priced off that bond, identifying it comes first.

For each candidate the short compares the cost of buying the bond against what it receives on delivery. Ignoring carry, the relevant quantity is the **gross basis**:

$$\text{gross basis} = \text{quoted bond price} - \text{futures price} \times \text{conversion factor} \tag{6.6}$$

and the **cheapest-to-deliver** (CTD) bond is the one that minimizes it.

Example 6.5 Identifying the cheapest-to-deliver bond

The futures price is 110.50 and three bonds are deliverable:

Bond	Quoted price	Conversion factor	Gross basis
A	98.50	0.8912	0.02240
B	104.25	0.9430	0.04850
C	112.75	1.0195	0.09525

For bond A, $98.50 - 110.50 \times 0.8912 = 98.50 - 98.4776 = 0.0224$; the others follow the same way. Bond A is cheapest to deliver, at a cost 2.24 cents per \$100 of face above the futures invoice.

The differences look trivial (seven hundredths of a point separates A from C), but on the \$100 million position of Example 6.8 the gap between choosing correctly and incorrectly is \$72,850.

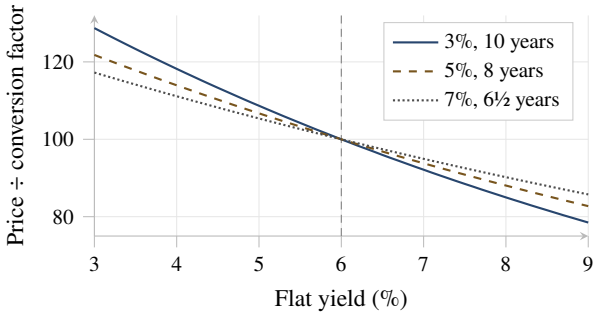


Figure 6.2: Price divided by conversion factor for three deliverable bonds of the ten-year basket, at a flat yield.

Rules of thumb. Two regularities recur. When yields are above the 6% notional coupon, low-coupon long-maturity bonds tend to be cheapest to deliver; when yields are below 6%, high-coupon short-maturity bonds tend to be. The reason is duration. A factor fixes each bond's value as if it yielded 6%; when yields are higher, every bond is worth less than that, and the long-duration bonds fall furthest below their factor, while when yields are lower every bond is worth more, and the short-duration bonds rise least above it. Since yields have been below

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6% for most of the contract's recent history, the CTD has usually sat at the short end of the basket. Figure 6.2 draws this for three bonds of the ten-year basket, the middle one being the 5% bond of Example 6.3. Divided by its conversion factor, each bond's price is exactly 100 at a 6% yield, so the three lines cross there, and the cheapest to deliver, the lowest line, switches from the short high-coupon bond to the long low-coupon one as yields rise through 6%. Near 6% the lines sit close together and a small move can change which is lowest, which is why the quality option described below is worth most when yields are near the notional coupon.

Carry and the net basis

The gross basis of (6.6) ignores what happens between now and delivery. Meanwhile the bond's holder earns its coupon and pays repo to finance it; the difference is **carry**:

$$\text{carry} = \text{coupon income over the holding period} - \text{repo financing cost} \tag{6.7}$$

and the quantity that actually measures the cost of the delivery option is the **net basis**:

$$\text{net basis} = \text{gross basis} - \text{carry} \tag{6.8}$$

Example 6.6 computes both for one deliverable bond.

Example 6.6 Carry and net basis

A deliverable bond carries a 4.5% coupon, is quoted at 101.20, and can be financed at a repo rate of 4.30% for the 90 days to delivery. Its gross basis is 0.0500 per \$100 of face.

Coupon income accrues on an actual/actual basis over 90 days:

$$4.50 \times \frac{90}{365} = 1.1096 \text{ per \$100 face}$$

Repo interest is charged at actual/360, here on the quoted price for simplicity (strictly it is charged on the cash price, accrued interest included):

$$101.20 \times 0.0430 \times \frac{90}{360} = 1.0879$$

so carry is $1.1096 - 1.0879 = +0.0217$, and from (6.8)

$$\text{net basis} = 0.0500 - 0.0217 = 0.0283$$

That 0.0283 is not a residual to be explained away: it is what the short's delivery options cost. A trader who buys this bond and sells futures against it acquires those options and pays 0.0283 per \$100 of face for them (0.028% of the cash price, or about 11 basis points annualized over the 90 days). The gross basis of 0.0500 confuses that price with the carry; the net basis separates them, and is the number a basis trader watches.

The sign of carry depends on whether the coupon exceeds the financing rate. In the low-rate decade carry was reliably positive and the distinction was a refinement; once short rates rose above the coupons on older bonds, carry turned negative for much of the deliverable basket, and net basis exceeded gross basis. Any statement about basis-trade profitability that ignores carry is incomplete, including the deliberately simplified Example 6.8 below.

The same ingredients give a theoretical futures price. Set the delivery options aside and treat the contract as a forward on the cheapest-to-deliver bond, which the short could buy today, finance in repo, and deliver. The bond's forward price is its cash price (quoted price plus accrued interest) carried to delivery at the repo rate, less the value of any coupon paid in between, as for an asset with known income (Section 5.3). Deducting the accrued interest the long will pay separately at delivery gives a quoted forward price, and dividing by the conversion factor turns that into a futures price, since the invoice of (6.4) pays the futures price times the factor. Example 6.7 carries it out.

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Example 6.7 The futures price from the cheapest-to-deliver bond

Take the bond of Example 6.6 as the cheapest to deliver, with a conversion factor of 0.9180. Its last coupon was 60 days ago, none falls in the 90 days to delivery, and accrued interest is counted as there.

Today's cash price is the quoted price plus 60 days of accrued interest:

$$101.20 + 4.50 \times \frac{60}{365} = 101.20 + 0.7397 = 101.9397$$

Carried to delivery at the 4.30% repo rate (actual/360), it becomes the bond's forward cash price:

$$101.9397 \times \left(1 + 0.0430 \times \frac{90}{360}\right) = 103.0356$$

At delivery the long pays 150 days of accrued interest separately, so the quoted forward price is

$$103.0356 - 4.50 \times \frac{150}{365} = 103.0356 - 1.8493 = 101.1863$$

and the theoretical futures price is

$$\frac{101.1863}{0.9180} = 110.2247$$

The gross basis of 0.0500 in Example 6.6 implies a market futures price of $(101.20 - 0.0500)/0.9180 = 110.1852$, which is 0.0395 lower. Multiplied by the conversion factor, the gap is 0.0363 per \$100 of face: the net basis of 0.0283, plus 0.0080 of repo interest on the accrued interest, which that example left out for simplicity. The futures contract trades below its forward-based value by what the short's delivery options are worth.

The delivery options

The short's right to choose which bond to deliver is a genuine option, and it is not the only one. Four are conventionally distinguished, all belonging to the short and all therefore depressing the futures price below what a no-arbitrage calculation on today's CTD alone would give.

The quality option is the one just described: the right to deliver any bond in the basket. Its value comes from the possibility that the CTD *changes* before delivery. Since the identity of the CTD depends on the level and shape of the curve, a large enough move switches it, and the short benefits from switching to whichever bond has become cheapest. The option is therefore worth more when the basket is wide and yields sit near the 6% notional coupon, where small moves can flip the ranking, and worth less when one bond is cheapest by a wide margin.

The timing option is the right to choose which business day of the delivery month to deliver. When carry is positive the short prefers to wait, earning more coupon than financing costs; when carry is negative it prefers to deliver early. Since the sign of carry is known, this option is worth little in practice—but it is not zero, because carry can change.

The wild card option arises from a mismatch in trading hours. The futures settlement price is struck at 2:00 p.m. Chicago time, but the short's notice of intention to deliver is not due until 6:00 p.m., while the cash market's New York session runs to 5:00 p.m. New York time (4:00 p.m. in Chicago). So the invoice amount is fixed for four hours, of which the first two have a liquid cash market to act in. If bond prices fall in that window, the short buys the deliverable cheaply and delivers it against the higher invoice already fixed at 2:00. It recurs every day of the delivery month, so a small daily edge adds up.

The end-of-month option extends the same idea: futures trading ends some days before the last delivery date, so for those days the invoice price is fixed while the cash market moves.

None of these is large individually. Collectively they are the reason a Treasury futures contract cannot be priced as a simple forward on the current cheapest-to-deliver bond, and the reason that basis traders speak of being short an option rather than holding an arbitrage.

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6.5 THE CASH–FUTURES BASIS TRADE

The relationship between a Treasury futures contract and its cheapest-to-deliver bond gives rise to a trade that has grown from an arbitrage niche into a staple of financial stability reports: buy the cash bond, sell futures against it, finance the bond in repo, and hold to delivery.

The profit is worth stating carefully, because the sign is easy to get wrong. The trade pays away the gross basis (you bought at the cash price and will receive the futures invoice) and earns the carry along the way. Its profit and loss (P&L) to delivery is therefore

$$\text{profit} = \text{carry} - \text{gross basis} = -\,\text{net basis} \tag{6.9}$$

using the net basis of Section 6.4. The trade is profitable precisely when the *net basis is negative*, which happens when the futures are rich relative to the cash bond, the condition that the asset-manager duration demand of Section 6.3 tends to create. Equivalently, and as Section 5.10 puts it, it is profitable when the implied repo rate exceeds the actual repo rate.

A *positive* net basis means the reverse: the cash-and-carry loses, and what it loses is the value of the short's delivery options from Section 6.4. That is the textbook state of affairs, and it is why the trade is not simply a riskless profit waiting in the market.

Two forces therefore set the sign, and they push opposite ways. The delivery options belong to the short and *depress* the futures price, driving the net basis positive. Demand for duration in futures form (from asset managers, and from liability-driven investors who want exposure without balance sheet) makes futures *rich*, driving it negative. Which dominates is an empirical matter, and since the 2010s the demand side has usually won in the liquid front contracts. That is what turned this from an occasional arbitrage into a standing business large enough to be a financial-stability question, and it is also why the trade is not riskless: the sign it depends on is set by a balance of flows, not by an identity.

Figure 6.3 sets out the three legs: the bond bought in the cash market, the futures sold against it, and the repo that finances the position until delivery closes all three.

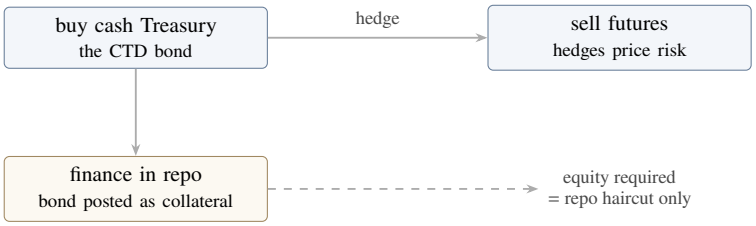


Figure 6.3: Structure of the cash–futures basis trade.

Example 6.8 Leverage in the basis trade

A fund buys \$100 million face of a cheapest-to-deliver bond at 98.50 and sells the corresponding futures. Asset-manager demand has left the futures rich, so the net basis is −0.0180 per \$100 of face. From (6.9) the position earns, held to delivery,

$$\$100{,}000{,}000 \times \frac{0.0180}{100} = \$18{,}000$$

On its own this is unimpressive: \$18,000 on a \$98.5 million position is 1.8 basis points. What transforms it is financing: the repo lender advances the bond's value less a haircut (Section 4.1), which the borrower funds itself. Treasury repo haircuts are small (take 2%), so the fund posts

$$0.02 \times \$98{,}500{,}000 = \$1{,}970{,}000$$

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of its own money and borrows the rest. The return on equity is

$$\frac{18{,}000}{1{,}970{,}000} = 0.91\%$$

over the life of the trade. If that is three months, the annualized return on equity is roughly 3.7%—from a position whose market risk is hedged.

The leverage is close to fifty times. At a 1% haircut it would be a hundred times.

Why it matters beyond the fund

Example 6.8 explains both why the trade is attractive and why regulators watch it. The economics work only at scale and with high leverage, and hedge funds' aggregate position has been estimated in the high hundreds of billions of dollars. Three features make that uncomfortable:

- **The leverage is procyclical.** Repo haircuts widen under stress. A move from 2% to 4% halves the position a given equity can support, forcing sales when markets are illiquid.
- **Margin on the futures leg moves the other way.** A sharp rally raises variation margin calls on the short futures position, and that margin must be met in cash even though the offsetting gain on the bond is unrealized. This is the mechanism of Chapter 8, appearing here.
- **Everyone is in the same trade.** The positions are concentrated in a small number of funds running similar strategies, so unwinding is correlated.

Case Study 6.2 March 2020

In the second week of March 2020, the US Treasury market (the world's deepest securities market, and the benchmark risk-free asset) stopped functioning properly. Bid–ask spreads widened by an order of magnitude, and on some days prices for on-the-run (most recently issued) and off-the-run (older) issues diverged without precedent.

The causes were several: a global dash for cash, foreign official selling, and dealer balance sheets already at their limits. A widely cited contributor was the forced unwinding of leveraged basis positions. As volatility spiked, haircuts rose and futures margin calls landed, and funds sold cash Treasuries into a market that could not absorb them, which pushed cash prices further from futures, widened the basis, and triggered more selling.

The Federal Reserve ended the episode by buying Treasuries on a scale not previously contemplated, roughly $1 trillion within weeks. The lesson is that a trade genuinely hedged against *price* risk can still fail catastrophically through *funding* risk, and leverage that looks prudent at normal haircuts does not survive stressed ones.

The clearing response

The principal regulatory response has been to push Treasury cash and repo activity into central clearing, under the SEC rules that Section 2.11 sets out with their compliance dates. Their intended effect is the multilateral netting of Example 2.4, together with uniform margin standards in place of bilaterally negotiated haircuts.

For this trade the rules bite through the financing leg. Once the repo that funds the position is cleared, its haircut comes from the clearing house's published margin model rather than from a negotiation with one dealer, and that puts a ceiling on the leverage in Example 6.8: a haircut that a dealer might set at 1% for a good client becomes a number the clearing house sets for everyone, and the example's fifty-to-a-hundred-times leverage needs a small haircut. The offsetting benefit is netting (a fund whose futures and repo positions are margined together

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posts against the net exposure, not the gross), and the practical question, unresolved as this is written, is which of the two effects dominates for a trade whose economics are a few basis points.

6.6 HEDGING A SWAP BOOK WITH THE STRIP

Section 6.2 named dealers hedging swap books as the largest source of SOFR futures volume. Chapter 7 treats swaps; only their outline is needed here. An interest rate swap exchanges a fixed rate for a floating one on a notional principal, period by period, and only the difference between the two payments changes hands. In each period the party that receives fixed is paid the fixed rate and pays the floating rate set for that period, here SOFR compounded over the quarter, so it gains when rates fall and loses when they rise. The party that pays fixed holds the opposite position.

For a dealer that has received fixed, the exposure is not a single number: each accrual period of the swap carries its own forward rate risk, and the futures strip lets those be hedged period by period. Example 6.9 hedges the first year of a two-year swap this way.

Example 6.9 Hedging the front of a swap

A dealer has received fixed on a \$500 million two-year swap with quarterly floating periods. Consider the first four accrual periods, each with $\tau = 0.25$.

A one basis point rise in any one period's forward rate costs the dealer about

$$\$500{,}000{,}000 \times 0.0001 \times 0.25 = \$12{,}500$$

before discounting. At \$25 per basis point per contract, the dealer sells

$$\frac{\$12{,}500}{\$25} = 500 \text{ contracts}$$

against each accrual period, 2,000 across the first year.

That match needs the dates to line up, and usually they do not. It assumes the swap's accrual periods coincide with the futures' reference periods, which holds only if the swap starts on one of the quarterly IMM dates (the third Wednesdays of March, June, September, and December, on which those reference periods begin and end). Suppose instead the first period runs from May 10 to August 10: 92 days, of which 39 fall before the third Wednesday of June and 53 after. The March contract covers the first stretch and the June contract the second, so the period's \$12,500 of sensitivity is split between them in proportion to days,

$$\frac{39}{92} \times \$12{,}500 = \$5{,}299 \to 212 \text{ March}, \qquad \frac{53}{92} \times \$12{,}500 = \$7{,}201 \to 288 \text{ June}$$

still 500 contracts, but spread across two delivery months. The residual is real: the two contracts do not move identically, so the hedge retains a small basis between the two reference periods, and a dealer whose book is mostly off-IMM swaps carries it on every one.

Three qualifications follow. Discounting reduces the required position slightly for later periods, since the \$12,500 arrives at the end of the accrual period rather than today. The futures hedge the futures rate, which by Section 30.3 exceeds the forward rate, so the hedge inherits convexity risk that must be managed separately. And as Figure 6.1 showed, each contract's sensitivity decays through its reference quarter while the swap's does not, so the ratio must be revisited within the period rather than only at the roll.

Beyond two years or so the strip thins out and dealers switch to hedging with other swaps or with Treasury futures, accepting the basis risk that comes with hedging a swap exposure using a government bond instrument. That basis, the swap spread of Section 7.9, is itself a traded risk.

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6.7 DURATION-BASED HEDGING

The most common use of Treasury futures is to adjust the interest rate exposure of a bond portfolio without trading the bonds themselves.

Chapter 4 gave the first-order relationship $\Delta B \approx -BD\Delta y$. Applying it to both the portfolio and the futures contract, and requiring the changes to offset, gives the number of contracts to short:

$$N^* = \frac{P\,D_P}{F\,D_F} \tag{6.10}$$

where P is the value of the portfolio, D_P its duration at the hedge horizon, F the contract value of one futures contract, and D_F the duration of the cheapest-to-deliver bond at contract maturity.

Example 6.10 Hedging a bond portfolio

A fund holds \$10 million of bonds with a duration of 6.8 years and wishes to hedge with ten-year note futures. The futures price is `110-16`, so with \$100,000 face per contract,

$$F = \frac{110.50}{100} \times \$100{,}000 = \$110{,}500$$

The cheapest-to-deliver bond has a duration of 7.6 years at contract maturity. From (6.10),

$$N^* = \frac{10{,}000{,}000 \times 6.8}{110{,}500 \times 7.6} = \frac{68{,}000{,}000}{839{,}800} = 80.97$$

so the fund shorts 81 contracts.

Check: a one basis point rise in yields costs the portfolio about $10{,}000{,}000 \times 6.8 \times 0.0001 = \$6{,}800$, while 81 contracts gain about $81 \times 110{,}500 \times 7.6 \times 0.0001 = \$6{,}802$.

What this hedge does not do. Equation (6.10) immunizes against a *parallel* shift only. A portfolio hedged this way remains exposed to changes in the shape of the curve (the bucketed exposures of Section 29.10) and to the possibility that the CTD bond changes, which alters D_F discontinuously. It also inherits the limitation of duration itself: it is a first-order approximation, and Chapter 4's convexity term reappears for large moves.

Hedging to a target duration

Managers more often want to *change* duration than to eliminate it. If the target duration is D_T rather than zero, the required position becomes

$$N^* = \frac{P\,(D_P - D_T)}{F\,D_F} \tag{6.11}$$

which is positive when duration must be reduced (short futures) and negative when it must be raised (long futures); Example 6.11 works the second case.

Example 6.11 Raising portfolio duration

The fund of Example 6.10 decides its \$10 million portfolio should carry a duration of 9.0 years rather than 6.8, because it expects yields to fall. From (6.11),

$$N^* = \frac{10{,}000{,}000\,(6.8 - 9.0)}{110{,}500 \times 7.6} = \frac{-22{,}000{,}000}{839{,}800} = -26.2$$

The negative sign means buying: the fund goes long 26 contracts.

The alternative, selling short-dated bonds and buying long ones, achieves the same duration but requires \$10 million of transactions, several days of execution in less liquid issues, and realized gains or losses on every position touched. The futures route requires margin on 26 contracts and

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takes seconds.

Futures contracts expire while the exposures they hedge often do not, so a five-year liability hedged with three-month contracts is stacked and rolled, as Section 3.9 describes: each roll crosses a bid–ask spread and is made at whatever spread between the two contracts prevails on the day, which is not known in advance. Section 3.10 tells the Metallgesellschaft episode, in which such a hedge failed through roll costs and margin calls while the hedged position performed as designed, the same funding and roll risk that recurs in Section 6.5.

6.8 BUILDING THE FRONT OF THE CURVE

Section 4.8 bootstrapped a zero curve from OIS quotes. In practice the region from roughly three months to five years is pinned down instead by the futures strip, because that is where the liquidity concentrates. The procedure has one wrinkle, and it is the subject of Chapter 30.

Each contract gives a futures rate for a specific three-month period. Converting the strip into forward rates requires subtracting the convexity adjustment of (6.3); only then can the forwards be chained into discount factors. Example 6.12 does so for three contracts.

Example 6.12 From futures quotes to forward rates

Three consecutive contracts settle at 95.72, 95.55, and 95.41, implying futures rates of 4.28%, 4.45%, and 4.59% for periods beginning in 1.00, 1.25, and 1.50 years. Take a short-rate volatility of 1.00% per year and apply the Ho–Lee adjustment $\frac{1}{2}\sigma^2 T_1 T_2$ from Section 30.3:

T_1	T_2	Futures rate	Adjustment	Forward rate
1.00	1.25	4.280%	0.63 bp	4.2738%
1.25	1.50	4.450%	0.94 bp	4.4406%
1.50	1.75	4.590%	1.31 bp	4.5769%

For the first contract, (6.3) gives $\frac{1}{2} \times 0.0001 \times 1.00 \times 1.25 = 0.0000625$, or 0.63 basis points; the others follow.

At this point on the curve the corrections are under a basis point and a curve builder might be tempted to skip them. The temptation should be resisted, because the adjustment grows with the square of maturity: by the eight-year contract it reaches 33 basis points.

Beyond the liquid futures region, the curve is completed with swap quotes, and the two must be made consistent; that is the simultaneous calibration described in Section 7.6.

SUMMARY

Interest rate futures come in two families: short-term contracts settle in cash against an overnight rate averaged over a reference period, and Treasury futures require delivery of a bond from a defined basket. Short-term contracts are quoted as $100 - R$, so a trader expecting higher rates sells; because the settlement rate accrues through the reference period, a contract's sensitivity decays toward zero across it, and hedges must be rebalanced within the quarter rather than rolled on one date. Bonds are quoted clean, with accrued interest added to reach the cash price.

Each contract's users explain much of its behavior. The SOFR strip serves borrowers hedging funding, dealers hedging the front of their swap books, and macro traders positioning on the policy path. Treasury futures serve asset managers and liability-driven investors moving duration, dealers hedging inventory, and mortgage holders rebalancing negative convexity, and the steady demand for duration in futures form tends to leave futures rich to the cash bond.

Conversion factors set at a 6% notional coupon make the basket only approximately equivalent, so one bond is cheapest to deliver and the contract tracks it, and which bond that is

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switches as yields cross the notional coupon. The futures price is that bond's forward price divided by its conversion factor, less the value of the short's options to choose the bond and the delivery day; the net basis, the gross basis less carry, is what those options cost.

The basis trade buys the cheapest bond, sells futures against it and finances it in repo. It earns a convergence of a few basis points, and thin repo haircuts make the leverage high, so it runs at scale and is fragile under stress. March 2020 showed that a trade hedged against price can still fail through funding, and central clearing of Treasury cash and repo is the main regulatory response.

Duration-based hedging sells $N^* = PD_P/(FD_F)$ contracts against a parallel shift and leaves curve reshaping, cheapest-to-deliver switches and convexity unhedged; a dealer instead matches the strip to a swap book period by period. Hedges that outlive their contracts must be rolled, which exposes the hedger to the roll spread every time. Finally, the strip anchors the front of the curve, once the convexity adjustment, negligible for near contracts and material for distant ones, is subtracted from the futures rates.

FURTHER READING

Burghardt and Belton's *The Treasury Bond Basis* is the standard practitioner treatment of conversion factors, the cheapest-to-deliver, and the delivery options, and it is the book to own if this material is your job. The Committee on the Global Financial System's *Repo Market Functioning* (2017) covers the financing leg that the basis trade depends on. On what happens when that financing withdraws, Duffie's Brookings paper on redesigning the Treasury market after March 2020 and the Financial Stability Board's thematic reviews of leverage in non-bank financial intermediation (2023–2024) are the two accounts to read together.

PRACTICE QUESTIONS

6.1 A Treasury bond is quoted at 104-20 with accrued interest of \$0.85 per \$100 of face. Express the quoted price as a decimal and give the cash price.

6.2 A trader buys 150 three-month SOFR futures at 96.10 and the contract settles at 96.34. Calculate the profit, state what happened to the implied rate, and explain why the contract is quoted as $100 - R$ in the first place.

6.3 Explain why the rate sensitivity of a three-month SOFR futures contract declines through its reference period, why this did not happen with Eurodollar futures, and what it implies for a hedger.

6.4 A bond pays a 6% coupon semiannually and has exactly 10 years to maturity after rounding. Calculate its conversion factor at the 6% notional coupon rate and explain the result. Then use equation (6.5) to calculate the conversion factor of a 4% coupon bond with 7 years and 3 months remaining after rounding.

6.5 The futures price is 108.75 and three bonds are deliverable, quoted at 96.20, 101.85, and 118.40, with conversion factors 0.8825, 0.9350, and 1.0880. Identify the cheapest-to-deliver bond.

6.6 Distinguish gross basis from net basis. A deliverable bond has a 3.0% coupon, is quoted at 94.60, can be financed at 4.40% for the 60 days to delivery, and has a gross basis of 0.0410. Calculate carry and the net basis, and comment on the sign.

6.7 A portfolio worth \$25 million has a duration of 5.4 years, ten-year note futures are priced at 109-00, and the CTD has a duration of 8.1 years at contract maturity. How many contracts should be shorted to immunize it? What position instead moves a \$40 million portfolio of duration 4.2 years to a target of 7.5 years, with futures at 111-00 and a CTD

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duration of 8.0?

6.8 Describe the cash–futures basis trade, and state where the profit comes from and where the leverage comes from.

6.9 A ten-year note futures contract settles at `112-08`. The delivered bond has a conversion factor of 0.9125 and accrued interest of \$0.60 per \$100. Calculate the invoice amount for one \$100,000 contract.

6.10 Explain why conversion factors do not make all deliverable bonds exactly equivalent, and state which bonds tend to be cheapest to deliver when yields are below 6% and why.

6.11 Explain why steady demand from asset managers for duration in futures form tends to leave Treasury futures rich relative to the cash bond, and connect this to Section 6.5.

6.12 A dealer has received fixed on a \$300 million swap with quarterly floating periods. How many three-month SOFR futures should it sell against each accrual period, ignoring discounting, and give two reasons the true hedge ratio differs.

6.13 Drawing on Sections 3.9 and 3.10, explain what a stack-and-roll hedge is and the two costs it incurs, and why a hedge of this kind can fail even when the underlying exposure behaves exactly as expected.

FURTHER QUESTIONS

6.14 A fund runs a \$500 million basis position at a net basis of -0.031 per \$100 with a repo haircut of 1.5%. Calculate the profit on full convergence, the equity required, the return on equity, and the leverage ratio. If the haircut widens to 4%, how much additional equity must the fund find, and what are its options if it cannot?

6.15 The basis trade is hedged against the price of Treasuries, yet it failed in March 2020. Explain precisely which risk was unhedged, identify the two distinct channels through which it materialized, and explain why both worsened simultaneously.

6.16 Explain why the short's delivery options depress the Treasury futures price. Would you expect the effect to be larger when the yield curve is flat or steep, and when yields are near 6% or far from it?

6.17 A dealer hedges a swaps book with SOFR futures. Explain three distinct reasons the hedge will not be exact, drawing on this chapter, Chapter 7, and Chapter 30.

6.18 Central clearing of Treasury cash and repo is intended to reduce systemic risk. Set out the mechanisms by which it should do so, and then argue the opposite case as strongly as you can. What would you want to know about the clearing house before deciding?

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